

CONTENTS

1. 🌱 Kinetic Energy
2. 🌱 Potential Energy
3. 🌱 Pressure
4. 🌱 Temperature
5. 🌱 Volume
6. 🌱 Conservation Of Energy
7. 🌱 Heat Transfer
8. 🌱 Internal Energy
9. 🌱 Thermodynamic Work
- 10. 🌱 First Law Of Thermodynamics**
11. 🌸 Human Metabolism

Generated by SPL

First Law Of Thermodynamics

First Law Of Thermodynamics

Definition. The first law of thermodynamics is a statement of energy conservation applied to thermodynamic systems. It asserts that the change in the internal energy ΔU of a system equals the heat Q added to the system minus the work W done by the system on its surroundings:

$$\Delta U = Q - W$$

Here, $Q > 0$ when heat flows into the system, $W > 0$ when the system does positive work on the environment, and ΔU represents the net change in all microscopic kinetic and potential energies of the constituent

particles. This law rules out perpetual motion machines of the first kind — devices that produce work without an energy source.

Worked Example. An ideal gas enclosed in a cylinder absorbs $Q = 500 \text{ J}$ of heat from a reservoir while expanding quasi-statically against a constant external pressure of $P = 1.0 \times 10^5 \text{ Pa}$, increasing its volume by $\Delta V = 3.0 \times 10^{-3} \text{ m}^3$. The work done by the gas is:

$$W = P \Delta V = (1.0 \times 10^5 \text{ Pa})(3.0 \times 10^{-3} \text{ m}^3) = 300 \text{ J}$$

Applying the first law:

$$\Delta U = Q - W = 500 \text{ J} - 300 \text{ J} = 200 \text{ J}$$

The internal energy of the gas increases by 200 J, consistent with a temperature rise in the gas.

Key Theorem. For any cyclic process, the system returns to its initial state, so $\Delta U = 0$. Consequently:

$$Q_{\text{net}} = W_{\text{net}}$$

The net heat absorbed equals the net work done. This result underlies the analysis of heat engines and refrigerators, where one evaluates thermal efficiency $\eta = W_{\text{net}}/Q_{\text{in}}$ over a complete cycle — the single new quantity introduced here to characterize how effectively a cyclic device converts heat into useful work.

Lab Exercise. Using a calorimeter equipped with an electrical heater and a calibrated piston-cylinder assembly, supply a measured quantity of electrical energy Q (in joules) to a fixed mass of gas. Record the piston displacement to compute $W = P \Delta V$. Calculate $\Delta U = Q - W$ and verify the result against the temperature change ΔT using $\Delta U = nC_v\Delta T$, where n is the number of moles and C_v is the molar heat capacity at constant volume (both treated as known quantities). Repeat the measurement at two or three different values of supplied energy to confirm that ΔU scales linearly with $Q - W$ and that the efficiency η remains consistent across trials.

